

7	(a) <i>either</i> if light passes through suitable film / cork dust etc. diffraction occurs and similar pattern observed or concentric circles are evidence of diffraction diffraction is a wave property	M1 A1 (M1) (A1)	[2]
	(b) (speed increases so) momentum increases $\lambda = h/p$ so λ decreases hence radii decrease <i>(special case: wavelength decreases so radii decreases – scores 1/3)</i> or (speed increases so) energy increases $\lambda = h / \sqrt{(2Em)}$ so λ decreases hence radii decrease	M1 M1 A1 (B1) (M1) (A1)	[3]
	(c) electron and proton have same (kinetic) energy <i>either</i> $E = p^2 / 2m$ or $p = \sqrt{(2Em)}$ ratio = $p_e / p_p = \sqrt{(m_e / m_p)}$ $= \sqrt{\{(9.1 \times 10^{-31}) / (1.67 \times 10^{-27})\}}$ $= 2.3 \times 10^{-2}$	C1 C1 C1 A1	[4]